

## Demonstration of Proposed Features

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Date</b>          |                                                   |
| <b>Team ID</b>       |                                                   |
| <b>Project Name</b>  | Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 1 Mark                                            |

### Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the ability to deliver on the proposed solution.

| S.No | Feature Name                          | Description                                                                                                 | Status<br>(Implemented/Partial/Pending) | Demonstrated<br>(Yes/No) | Remarks                                                                                     |
|------|---------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------|---------------------------------------------------------------------------------------------|
| 1    | AI-Based Crop Recommendation          | Predicts the most suitable crop from soil and climate parameters using a trained Logistic Regression model. | Implemented                             | Yes                      | Achieves 97.3% test accuracy across 22 crop classes                                         |
| 2    | Web-Based User Interface              | Flask web application with Home, About, Find Your Crop, and Result pages.                                   | Implemented                             | Yes                      | Demonstrated live during the walkthrough                                                    |
| 3    | Input Validation & Feature Scaling    | Validates entered values and applies the fitted StandardScaler before prediction.                           | Implemented                             | Yes                      | Verified with multiple input value sets                                                     |
| 4    | Public Deployment                     | Application hosted and accessible via a public URL without installation.                                    | Implemented                             | Yes                      | Live at <a href="https://opticrop-407a.onrender.com">https://opticrop-407a.onrender.com</a> |
| 5    | Exploratory Crop Clustering (K-Means) | Groups crops by environmental profile to study natural clusters.                                            | Partial                                 | No                       | Used for research/EDA only; not integrated into the live prediction flow                    |
| 6    | Fertilizer & Multilingual Support     | Fertilizer suggestions and crop recommendations in multiple languages.                                      | Pending                                 | No                       | Planned as a future enhancement, not yet implemented                                        |

### Feature Implementation Summary:

|                                        |                                |
|----------------------------------------|--------------------------------|
| <b>Total Features Proposed</b>         | 6                              |
| <b>Total Features Implemented</b>      | 4                              |
| <b>Total Features Demonstrated</b>     | 4                              |
| <b>Overall Implementation Rate (%)</b> | 67% (4 of 6 fully implemented) |